Antiperspirant (Al-salt, dirty sibling of r9 deodorant): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for antiperspirant (Al-salt, dirty sibling of r9 deodorant) in Cosmetics and Personal Care — Leave-on, evaluated on finished leave-on cosmetic, TOTAL CONTENT as placed on market (ug/kg of product as sold), on the technically--unavoidable-impurity basis every instrument in this family uses. Not a migration/leachate basis, not a dietary basis, not an ingredient purity specification.. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0, government-limit-only initial setting: every numeric cell is the strictest applicable sovereign maximum for this exact leave-on cosmetic form expressed as total content of the finished product as placed on market, or a disclosed read-across where the exact cell is unregulated, or a substantiated not-material determination, or a control. No cell is occurrence-derived, no cell is precautionary or feasibility-adjusted, and no cell is anchored on an ingredient or colour-additive purity specification.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS

Lead (Pb)	2,000	government maximum level
Cadmium (Cd)	100	government maximum level
Total arsenic (tAs)	500	government maximum level
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	100	government maximum level
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	10,000	government maximum level
Tin (Sn)	3,750,000	human-grade read-across (European Union)
Aluminium (Al)	n/m	not material in this matrix
Total chromium (Cr)	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	5	human-grade read-across (European Union)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	2,000	government maximum level
Cadmium (Cd)	100	government maximum level
Total arsenic (tAs)	500	government maximum level
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	100	government maximum level
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	10,000	government maximum level

Tin (Sn)	3,750,000	European Union — human-food analog
Aluminium (Al)	n/m	Aluminium is the PERMITTED ACTIVE INGREDIENT of an antiperspirant, not a contaminant of it. Aluminium chlorohydrate, aluminium chloride, aluminium sesquichlorohydrate and aluminium zirconium tetrachlorohydrex glycine are the astringent actives that make the product work; the US over-the-counter antiperspirant monograph at 21 CFR 350 lists them as permitted actives and regulates their concentration as an active-ingredient specification expressed in percent of the anhydrous salt, and the EU restricts the aluminium zirconium complexes as ingredients under Regulation 1223/2009 Annex III. Neither is an impurity ceiling. Publishing a total-aluminium contaminant maximum for this form would be a category error: every compliant product would exceed any contaminant-scale value by three to four orders of magnitude, so the number would police the label claim rather than any contamination, and a brand could satisfy it only by removing the active. No credible sovereign publishes an aluminium contaminant maximum for an antiperspirant or for any other cosmetic form.
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	5	European Union — human-food analog

4 | Cite this standard

Pendergrass, K. (2026). *Antiperspirant (Al-salt, dirty sibling of r9 deodorant): HMTc standard*. Institute of Contaminant Standards.

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References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Sn, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/antiperspirant. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

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